

RESEARCH ARTICLE

Artificial Intelligence Innovation, Firm Performance, and Market Perception: Does Sustainability Matter?

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ABSTRACT

This study analyzes how artificial intelligence (AI) innovation affects firms' financial performance and value among S&P 500 companies in the US market, using a panel data regression model for the period 2000–2024. The study finds mixed evidence of a small positive impact of AI innovation on firms' return on assets, with the effect stronger for more R&D-intensive firms. This study also tests whether the market considers the environmental impact of AI innovation when valuing a firm's AI innovation. The findings demonstrate that AI innovation increases firm value for companies that perform well on sustainability-related metrics. This effect is stronger under Democratic administrations than under Republican governments, indicating the influence of political environments on the market's perception of firms' actions. The results remain robust to multiple checks, including system-GMM and PSM. The study makes a compelling argument for incorporating sustainability as a crucial strategic component in the new AI-driven business world, while providing insights for policymakers, executives, and investors.

1 | Introduction

The global AI market, valued at USD 189 billion in 2023, is expected to increase 25-fold over the next 10 years, reaching USD 4.8 trillion by 2033 (United Nations Conference on Trade and Development (UNCTAD) 2025). AI is assuming increasingly important roles in business operations like customer service, inventory management, and process automation (Haan and Watts 2023). AI-based technologies have helped firms enhance productivity and improve process efficiencies (Aghion et al. 2017; Agrawal et al. 2019; Czarnitzki et al. 2023; Dirican 2015). Corporate strategists view AI as a critical component of their future success, given that nearly 50% of strategic planning and execution activities are expected to be automated (Gartner Inc. 2023). Situating AI innovations in firm and industry-specific scenarios can help firms develop a cost-effective, sustainable, competitive advantage (Kemp 2024). The adoption of AI can lead to the emergence of related complementarity and substitution dynamics, rendering conventional

competitive advantages obsolete (Krakowski et al. 2023). This study begins by analyzing whether AI innovation, measured as the number of AI-related patents a firm is granted in a year, translates into profitability by improving its return on assets (RoA). Existing literature mostly focuses on how AI impacts firm-level productivity (Damioli et al. 2021; Duan et al. 2023; Zhai and Liu 2023). Since AI innovation requires significant investments, it is necessary to analyze whether firms receive tangible financial benefits from it. Consequently, our study is the first to examine the direct impact of AI innovation on firm profitability. The USA is at the forefront of AI research and innovation, making it the ideal market to base our study. Since some studies have also found that AI adoption can negatively affect firm value due to high implementation costs (Lui et al. 2022), we aim to investigate whether large firms with sufficient buffers for AI investments can achieve better financial performance from these technologies. Hence, in the US market, we select firms in the S&P 500 as of December 31, 2024, as our sample. These are the leading 500 US public

companies, collectively accounting for approximately 80% of the US equity market capitalization (S&P Global), with their market performance thoroughly monitored by investors. Our analysis finds AI innovation to enhance the RoA of a firm, the effect being stronger for firms that are more R&D (research and development) intensive, providing evidence of the financial viability of investing in AI innovation and complementary dynamic organizational capabilities to properly absorb such innovation into firm operations.

This study then examines the role of sustainability in the age of rising AI innovation, particularly given its substantial environmental impact. The rise of AI has led to an increase in the number of data centers from 500,000 in 2012 to 8,000,000 in 2024 (United Nations Environment Programme 2024). This exponential rise comes with an estimated 120 million metric tons of e-waste by 2050 (World Economic Forum 2019), a huge strain on freshwater resources (P. Li, Yang, et al. 2023), and an expansive carbon footprint. Training an AI model can emit up to 626,000 pounds of carbon dioxide, 57 times the carbon footprint of an average human, and 5 times the emissions of an average car in its lifetime (Strubell et al. 2019). The energy consumption associated with developing AI technologies is expected to increase exponentially as they become more commonplace. Thus, this study analyzes the impact of AI innovation on firm value, based on overall sustainable performance.

Focusing on sustainability in firm operations is becoming a significant strategic decision for firms, especially considering the global policy push towards sustainable development (IPCC 2022; OECD 2012; United Nations 2015). More than 190 nations have also adopted recommendations on the ethical and environmentally sustainable application of AI (UNESCO 2021). It is now becoming apparent that focusing on sustainability while developing or adopting AI is not only a moral but a strategic imperative (Ren and Wierman 2024). Hence, we move on to analyze whether AI innovation by a sustainable firm is viewed differently from innovation by a non-sustainable firm.

Studies have already demonstrated the financial benefits of adopting sustainable business practices (Stefan and Paul 2008) and observed a positive link between sustainable performance and firm value (Aydoğmuş et al. 2022; Laskar and Gopal Maji 2018; Tahmid et al. 2022; Yu and Zhao 2015). Existing literature mostly focuses on either the environmental cost of AI (Strubell et al. 2019; Zhu et al. 2023) or on the positive impact of AI in terms of resource efficiency (Li et al. 2022; J. Li, Ma, et al. 2023). Thus, we analyze if the market perception of AI innovation differs depending on the sustainable performance of the firm, making our study the first to address the relevance of firm-level sustainability while evaluating the impact of AI innovation on firm value. AI innovation by a sustainable firm might lead investors to believe that (1) the AI-related technology is being developed in an energy-efficient manner or (2) the AI technology will be utilized sustainably by the firm, reducing its carbon footprint (Hassanein et al. 2024), or (3) the firm can offset the excessive energy and resource requirement of AI innovation due to its sustainable policies. In any of these cases, the firm should be viewed positively by the market. Using the composite environmental, social, and governance

(ESG) and the independent environmental pillar score as our sustainability metrics, we find, in a subsample analysis, that AI innovation by sustainable firms leads to an increase in firm value, whereas firms that fare poorly on sustainability receive no significant reward to their market value. The effect is even more pronounced in the period following the 2015 Paris Agreement, with sustainability attracting growing attention worldwide.

Sustainable performance is also observed to be affected by cultural and governance-related contexts (Hassanein and Elmaghrabi 2025; Hassanein et al. 2024; Elmaghrabi et al. 2025). Building on this premise, we examine the relevance of sustainability in the market perception for firm-level AI innovation in Democratic and Republican administrations due to their divergent views on sustainability (Jones 2023). The analysis further reveals that this effect is significant during the years of Democratic leadership in the United States, highlighting the difference in attitudes towards environmental issues based on political affiliation (Smith et al. 2024).

This study significantly advances the literature across several important dimensions. Empirically, it provides evidence of the benefits of investing in AI and in aligned organizational capabilities (R&D investment), thereby validating the strategic decision by firms to incur such significant expenses. AI innovation acts as a valuable, inimitable resource, providing firms with a sustained competitive advantage. This is the first study to shed light on the role of sustainability as we move towards the AI-driven future, with empirical evidence of sustainable firms receiving improved firm valuations for their AI innovations. It focuses on market perceptions of the environmental costs associated with AI and why adopting a combination of technological advancements and sustainable growth can be a rewarding strategic choice. These results align more with, and provide evidence for, the value-enhancing theory, which posits that improved sustainable performance can help firms improve their market value (Hart and Milstein 2003), over the shareholder theory, which prioritizes profit maximization over social responsibility (Friedman 1970), enriching the theoretical contributions of the study. On the policy front, this study is also the first to examine how governmental support for sustainable initiatives affects firm value, depending on policy-level support for sustainable initiatives, aligning the results with the perspectives of institutional theory (North 1990). Our study has practical implications for policymakers, managers, and investors. The findings suggest that, rather than merely a moral commitment, sustainable performance is emerging as a driver of firms' market value. Business executives can strategically disclose their ESG and energy-efficient AI initiatives as signals of long-term quality and market resilience, thereby translating sustainable performance into tangible financial returns and enduring competitive advantage. Policymakers can design incentives targeted towards sustainable technology development, such as tax rebates and subsidies for firms with strong ESG performance, and by funding more research in energy-efficient AI and green computing infrastructure (Bolón-Canedo et al. 2024). A positive market perception of sustainable firms increases the likelihood of success for such policies, thereby reducing enforcement costs. Investors can incorporate ESG-based AI metrics into their portfolio evaluation

to make smarter, more efficient, and more profitable investment choices, helping maximize long-term capital gains while reducing exposure to transition risk.

The remainder of this paper is structured as follows. The review of relevant background literature and the development of hypotheses are covered in Section 2. Section 3 provides details on the data sources for all variables, along with the methodology used to analyze them. Section 4 explains the empirical results, while robustness checks and alternative analysis techniques are presented in Section 5. Section 6 discusses the theoretical contributions, limitations, and future research scope of this study. Section 7 concludes our discussion.

2 | Literature Review and Hypotheses Development

2.1 | Artificial Intelligence and Firm Performance

The term “artificial intelligence” was first used by John McCarthy in a workshop at Dartmouth in 1956. Since the 1990s, AI has advanced at a considerable pace, reaching the current stage of Artificial General Intelligence, which utilizes deep learning and Big Data analytics to handle increasingly complex problems like loan underwriting, fintech, and robo-advising (Chen et al. 2019; D’Acunto et al. 2019; Fuster et al. 2022; Smith et al. 2006). Multiple studies have explored how different aspects of firm performance are influenced by AI. Industrial robot applications have been found to improve firm productivity significantly (Duan et al. 2023). Yang (2022) observes that AI innovation in the Taiwanese market leads to higher firm productivity, accompanied by a reduction in the labor force with lower educational qualifications. Increased productivity was also observed in German firms after the adoption of AI technologies (Czarnitzki et al. 2023). In a sample of Chinese listed firms, AI adoption enhanced a firm’s total factor productivity through reduced costs, more efficient labor utilization, and increased innovation, with the effect more pronounced for larger firms, state-owned enterprises, and labor-intensive industries (Zhai and Liu 2023). A similar increase in productivity following AI adoption was also observed in small- and medium-sized enterprises and service-sector firms (Damioli et al. 2021). Firms already investing in synergistic technologies are observed to extract more productive use of AI after incorporating it into their business operations (Calvino and Fontanelli 2023).

On the other hand, certain studies have also found firm performance to remain unaffected by AI (Huang and Lin 2025), while some have observed AI adoption to cause a reduction in firm value (Lui et al. 2022) due to high implementation costs. AI technologies have significant potential to transform business operations, resulting in reduced costs, increased efficiency, minimized human errors, and accelerated processes. Despite the well-documented operational benefits, there is limited research examining whether AI innovation translates into firm-level financial profitability. This leads us to our first research question, which examines whether AI innovation improves firm profitability, measured by RoA, among S&P 500 firms. Thus, our first hypothesis is as follows:

H1. *AI innovation positively impacts the profitability (RoA) of S&P 500 firms.*

Complementarities research suggests that technological innovation leads to enhanced firm performance when it is combined with aligned organizational capabilities (Torrent-Sellens et al. 2025a, 2023). Within the technology–organization–environment framework (Tornatzky et al. 1990), there is a need for sufficient internal innovation and absorptive capacity to successfully incorporate AI technologies into firm-level operations. In this study, we select R&D intensity as one such complementary capability. Research has established the importance of R&D as a critical component of the innovation ecosystem of a firm (Díaz-Chao et al. 2021; Camiña et al. 2020), which leads to an improvement in its technological absorption capability (Doraszelski and Jaumandreu 2013; Luintel et al. 2014). Thus, the profitability gains through AI innovation are expected to be stronger for firms with higher R&D intensity. This leads us to the second hypothesis of our study:

H2. *R&D intensity moderates the relationship between AI innovation and profitability (RoA) of S&P 500 firms.*

2.2 | Sustainability, Artificial Intelligence, and Firm Value

Research interest in sustainability and sustainable finance has been increasing rapidly (Benameur et al. 2024; Benameur et al. 2025). In the ongoing efforts to combat climate change, firms are now expected to engage more in sustainable business practices, with such performance being rewarded by the market. An improved ESG (environmental, social, and governance) score is linked to better firm value and performance (Aydoğmuş et al. 2022; Tahmid et al. 2022). In their systematic literature review on this topic, X. Li et al. (2024) find that most studies link better ESG performance to increases in a firm’s market value, highlighting the rising importance of sustainable business practices in investors’ and markets’ eyes. Environmental and social disclosure by firms in Europe and India has been observed to enhance their valuations and help them reduce policy-induced uncertainty (Ahsan and Qureshi 2021; Laskar and Gopal Maji 2018). Small- and medium-sized enterprises also reported improved performance due to green innovation (Schrank and Kijkasiwat 2024). Factors like gender diversity, earnings management, and corporate innovation also affect firm-level sustainable performance (Alazzani et al. 2017; Tahat and Hassanein 2024; Abdelbaky et al. 2025; Hassanein and Elsayed 2025). Sustainability is steadily emerging as a major corporate strategy, with greater managerial participation in such incentives enhancing a firm’s market reputation (Stefan and Paul 2008). A positive relationship, aligned with value-enhancing theory rather than shareholder value theory, has been established between sustainable performance and firm value. This positive effect is stronger in countries with high disclosure levels and strong investor protection (Yu and Zhao 2015).

AI innovation carries both potential environmental benefits and concerns. The computing power required for training,

operating, and updating AI doubles every 100 days, with projections indicating a millionfold surge over the following five years (Zhu et al. 2023). Tech giants like Microsoft and Google have reported that their greenhouse gas emissions have risen by 30% and 50%, respectively, over the past 4 years, primarily due to the expansion of data centers to sustain their AI infrastructure (Google 2024; Kimball 2024). Conversely, AI utilization through industrial robots has reduced carbon intensity and improved energy efficiency across multiple industrial sectors (Y. Li et al. 2022; J. Li, Ma, et al. 2023). Hussain et al. (2024) demonstrated that firms with higher AI adoption exhibit greater green innovation, with the effect more pronounced for private enterprises. AI has improved natural resource utilization efficiency (Wang et al. 2023), while facilitating the development of renewable energy across nations (Zhao et al. 2024), and helping firms achieve their sustainability-related metrics (Torrent-Sellens et al. 2025b). AI technology, if properly harnessed, is expected to potentially reduce 5%–10% of global greenhouse gas emissions (Kemene et al. 2024) and revolutionize the approach to sustainability in the financial sector (Hassanein and Tharwat 2024).

The presence of such mixed results opens up the avenue to investigate whether AI innovation influences firm value differently based on the firm's sustainability score. No existing study has explored the relevance of sustainable performance to market perceptions of a firm's AI innovation. Based on prior research, we propose that AI innovation in firms with sustainable performance should be viewed more positively by the market than innovation in firms that do not fare as well. This leads us to the third hypothesis of our study, presented below:

H3a. *AI innovation leads to an increase in firm value for firms with better sustainability-related outcomes.*

H3b. *AI innovation leads to a decrease/no effect in firm value for firms with poorer sustainability-related outcomes.*

2.3 | Sustainability and the Political Environment

Moving forward, firm-level sustainable performance is also significantly influenced by cultural, governance, and market-related contexts (Hassanein and Elmaghrabi 2025; Hassanein et al. 2024; Elmaghrabi et al. 2025). Institutional theory (North 1990) also suggests that the institutional environment (in the form of regulations, laws, and policies) influences firm outcomes and market valuations. This makes it reasonable to assume that market reactions to sustainability may differ depending on the ruling dispensation's viewpoint. Consequently, we expect political affiliation to significantly influence how the market perceives a firm's sustainability actions, particularly environmental sustainability. There has been an increasing divide between Democratic and Republican viewpoints concerning sustainability in the United States. Since the 1990s, Republicans have given lower priority to environmental issues than Democrats (E. K. Smith et al. 2024). 78% of Democrats prioritize the environment over economic growth, while only 20% of Republicans share this sentiment (Jones 2023). The percentage

of Democratic voters who believe the top priority for a President should be “Protecting the Environment,” “Global Warming,” and “Climate Change” is 85%, 56%, and 78%, respectively. The corresponding figures for the Republican voters are 39%, 16%, and 21%, respectively (Kennedy and Johnson 2020). Acknowledging the distinct partisan disparity in matters related to environmental sustainability, we are the first study to investigate whether AI innovation in sustainable firms is evaluated positively by the financial market during the more sustainability-oriented Democratic administrations compared to Republican administrations. Hence, we formulate the following fourth hypothesis:

H4a. *AI innovation leads to an increase in firm value for firms with better sustainability-related outcomes during Democratic administrations.*

H4b. *AI innovation does not significantly affect firm value for firms with better sustainability-related outcomes during Republican administrations.*

3 | Data and Methodology

3.1 | Dependent Variables—Return on Assets (RoA) and Tobin's Q

This study examines how AI innovation affects the performance and value of S&P 500 firms, utilizing annual data from 2000 to 2024. The S&P 500 comprises the 500 largest publicly traded firms in the US equity market, accounting for approximately 80% of the total US market capitalization. It is considered one of the best indicators of the US financial market and broader global market trends. These firms also lead in innovation in their respective sectors, with substantial investments in research and development. S&P 500 firms also have approximately 80% institutional ownership (Greenspon 2019). Institutional investors are considered more capable monitors of firm performance (Shleifer and Vishny 1986) and can ensure higher disclosure quality (Haq et al. 2024). Thus, the market valuation effects for these firms would be most measurable and meaningful. All these factors make them an ideal sample to determine whether AI innovation enhances their financial performance and firm value. Following prior literature, we have excluded financial sector firms from our study due to differences in their capital structure, accounting practices, and regulatory requirements compared to other firms (Poulsen et al. 2013; Tahat and Hassanein 2024). The dependent variable selected for the first hypothesis is return on assets (RoA). RoA is useful for understanding how efficiently a firm utilizes its resources to generate profits. It is also a widely accepted measure of firm financial performance (Coles and Li 2023; Dang et al. 2018; Mehran 1995). The annual return on assets for a firm i in year t ($RoA_{i,t}$) is calculated as follows:

$$RoA_{i,t} = \text{Net Income}_{i,t} / [(\text{Total Assets}_{i,t-1} + \text{Total Assets}_{i,t}) / 2] \dots \quad (1)$$

For measuring firm value, we use Tobin's Q as our primary dependent variable. It is one of the most used indicators of firm value in finance research (Carter et al. 2017). The annual Tobin's Q for a firm i in year t (Tobin's $Q_{i,t}$) is calculated as follows:

$$\text{Tobin's } Q_{i,t} = (\text{Market Value of Equity}_{i,t} + \text{Total Assets}_{i,t} - \text{Book Value of Equity}_{i,t}) / \text{Total Assets}_{i,t} \quad (2)$$

We winsorize both variables at the 1st percentile to account for extreme values. The data for both these variables is sourced from LSEG (London Stock Exchange Group) Workspace.

3.2 | Explanatory Variable—AI Patents

Following prior literature, we measure AI innovation by the number of AI-related patents granted to S&P 500 firms (Ménier 2022; Parteka and Kordalska 2023; Tseng and Ting 2013; USPTO 2020; Y. Wang and Liu 2025; Zhai and Liu 2023). Patents are considered a reliable proxy for AI innovation since firms are highly incentivized to patent any AI technology developed by them to prevent imitation and build a competitive advantage (Yang 2022). We utilize the methodology outlined in the PATENTSCOPE Artificial Intelligence Index to identify such patents. This index utilizes a combination of keywords, Cooperative Patent Classification (CPC), and International Patent Classification (IPC) symbols to identify patents related to AI technology. These patents cover a broad spectrum of AI techniques (machine learning, neural networks, deep learning, etc.), functional applications (augmented reality, predictive analytics, natural language processing, etc.), and application fields (business, biomechanics, telecommunications, etc.). The full list of keywords and patent classification symbols used is publicly available on the World Intellectual Property Organization (WIPO) website. The AI-patent database is then filtered using the firms' common names to identify patent ownership for S&P 500 firms.

3.3 | Other Variables

3.3.1 | Control Variables

Control variables at the firm level include—firm size (natural logarithm of total assets), cash holdings (cash and equivalents divided by total assets), leverage (total debt divided by total assets), profitability (operating income divided by total assets), R&D intensity (research and development expenses divided by total assets), return on equity, and analyst following (count of analysts covering the firm). The lagged values of dependent variables are also included in the model. The data for these variables is sourced from the LSEG Workspace.

3.3.2 | Sustainability-Related Variables

Sustainability filters include the composite ESG score as well as the separate Environmental pillar score for each firm. This data is sourced from the LSEG Workspace. The earliest available annual ESG data for our sample dates back to 2000, so we have selected the period from 2000 to 2024 for our study.

3.3.3 | Dummy Variables

We also incorporate certain dummy variables in our analysis. A COVID dummy that takes a value of 1 for 2020–2022, when the pandemic was prevalent globally, and 0 otherwise. Similarly, there are dummy variables for the dot-com bubble bust and the

Great Recession, which take the value 1 for the years 2000–2002 and 2008–2009, respectively. Investigating the influence of different political administrations on firm value, the sample will be analyzed under both Democratic and Republican presidencies in a subsample analysis. The Democratic presidencies include the periods of 2000 (under President Bill Clinton), 2009–2016 (under President Barack Obama), and 2021–2024 (under President Joe Biden). Similarly, the Republican presidencies include the periods of 2001–2008 (under President George W. Bush) and 2017–2020 (under President Donald Trump). All the aforementioned variables and their definitions are listed in Table 1.

3.4 | Methodology

The panel regression model examining the impact of AI innovation on firm performance (measured via RoA) is specified below:

$$RoA_{i,t} = \alpha + \beta_1 AI\ Patents_{i,t-1} + \beta_2 CONTROLS_{i,t-1} + \beta_3 Dummies_t + \beta_4 RoA_{i,t-1} + \varepsilon_{i,t} \quad (3)$$

TABLE 1 | List of variables used in the study.

Variable	Definition
$RoA_{i,t}$	The return on assets for firm i in year t
Tobin's $Q_{i,t}$	Tobin's Q for firm i in year t
$AI\ Patents_{i,t}$	The number of AI-related patents granted to firm i in year t
$Cash\ Holdings_{i,t}$	Cash and cash equivalents divided by total assets for firm i in year t
$Leverage_{i,t}$	Total debt divided by total assets for firm i in year t
$Size_{i,t}$	Logarithm of total assets for firm i in year t
$Profitability_{i,t}$	Operating income divided by total assets for firm i in year t
$R\&D_{i,t}$	R&D expenditure divided by total assets for firm i in year t
$RoE_{i,t}$	Return on equity for firm i in year t
$Analyst_{i,t}$	Analyst following for firm i in year t
$ESG_{i,t}$	Composite ESG Score for firm i in year t
$EPS_{i,t}$	Environmental Pillar Score of the ESG Score for firm i in year t
COVID	Dummy variable for the COVID-19 pandemic with a value of 1 for years 2020–2022, 0 otherwise
DotCom	Dummy variable for the dot-com bubble burst with a value of 1 for years 2000–2002, 0 otherwise
GR	Dummy variable for the great financial recession with a value of 1 for years 2008–2009, 0 otherwise

Source: Authors' own compilation.

where $RoA_{i,t}$ is the return on assets for firm i in year t , and $AI\ Patents_{i,t-1}$ is the number of AI-related patents granted to firm i in year $t-1$. $CONTROLS$ include firm size, cash holdings, leverage, profitability, and R&D intensity. $DUMMIES$ include the dummy variables for COVID, the dot-com bubble bust, and the Great Recession. Both industry and time fixed effects are incorporated in our model. To understand the moderating impact of R&D intensity on the relationship between AI innovation and firm profitability, we utilize the following model:

$$RoA_{i,t} = \alpha + \beta_1 AI\ Patents_{i,t-1} + \beta_2 AI\ Patents * R\&DInt_{i,t-1} + \beta_3 CONTROLS_{i,t-1} + \beta_4 Dummies_t + \beta_5 RoA_{i,t-1} + \varepsilon_{i,t} \quad (4)$$

where $AI\ Patents * R\&DInt_{i,t-1}$ is the interaction of the AI-related patents and the R&D intensity of firm i in year $t-1$.

Similarly, the model chosen to assess how AI innovation affects firm value is defined as follows:

$$Tobin's\ Q_{i,t} = \alpha + \beta_1 AI\ Patents_{i,t-1} + \beta_2 CONTROLS_{i,t-1} + \beta_3 Dummies_t + \beta_4 Tobin's\ Q_{i,t-1} + \varepsilon_{i,t} \quad (5)$$

where Tobin's $Q_{i,t}$ is the measure of firm value for firm i in year t . $CONTROLS$ include firm size, cash holdings, leverage, profitability, R&D intensity, return on equity, and analyst following. $DUMMIES$ include the dummy variables for COVID, the dot-com bubble bust, and the Great Recession. Industry and time fixed effects are also incorporated in our model. Table 2 presents the summary statistics for our variables. On average, 18.57 AI-related patents are granted to a firm in a year. But there is extreme variation in innovation intensity, with a high standard deviation of 115.6604. This is understandable, as the S&P 500 comprises some of the largest tech firms that invest heavily in AI innovation, whereas many other firms may not. The correlation coefficient matrix is provided in Table 3. We now turn to the empirical findings from these models in the following section.

TABLE 2 | Descriptive statistics.

Variable	Observations	Mean	Std. dev.	Min	Max
RoA	11,456	0.0736804	0.1048159	−1.5045	1.5756
Tobin's Q	11,233	1.898519	2.289228	0.002427	40.16096
AI patents	12,575	18.57408	115.6604	0	2538
Cash holdings	11,610	0.1311063	0.1586705	0	0.9749436
Leverage	11,610	0.277571	0.2396879	0	4.051822
Size	11,610	16.36373	1.739823	7.545918	22.11026
Profitability	11,610	0.0995841	0.1191426	−2.674068	1.795008
R&D	11,610	0.0252828	0.0546214	0	1.209076
RoE	11,041	0.3260184	5.762186	−248.5	315.6
Analyst	10,309	18.56814	9.145227	1	86
ESG score	8943	53.03111	20.26366	0.6	95.16
Environmental pillar score (EPS)	8943	45.76451	29.35752	0	98.55

Source: Authors' own compilation.

TABLE 3 | Correlation coefficient matrix.

	AI patents	RoE	Size	Cash holdings	Leverage	R&D	Profitability	Analyst
AI patents	1							
RoE	0.0008	1						
Size	0.1695	0.0037	1					
Cash holdings	0.1175	−0.0025	−0.3195	1				
Leverage	−0.0313	0.0369	0.0876	−0.3039	1			
R&D	0.1432	−0.0331	−0.2554	0.5358	−0.1747	1		
Profitability	0.0571	0.0732	−0.2293	0.1256	−0.1611	−0.0401	1	
Analyst	0.2347	0.0067	0.4634	0.1633	−0.1276	0.1502	0.0808	1

Source: Authors' own compilation.

4 | Empirical Results

4.1 | AI Innovation and Firm Performance

The regression results of our model, defined in Equation 3, are reported in Table 4. Column (1) reports the results of the univariate analysis, while Column (2) incorporates the control variables, along with the time and industry fixed effects. The results show AI innovation has a significant positive impact on firm performance, with each new AI patent associated with a 0.002% rise in the next year's RoA. This suggests that, although the results may not be immediate, companies are investing in AI innovation to enhance their long-term competitive advantage and performance. Firm size hurts RoA, a result consistent with Mehran (1995). Leverage, R&D intensity, and profitability positively affect RoA, as these variables enable a firm to run its operations smoothly and use its resources efficiently. The dot-com bubble burst, and the Great Financial Crisis significantly hurt firm performance, with profits nosediving during these events. The positive coefficient of the COVID dummy might seem counterintuitive at first. However, this could be attributed to firms sharply increasing product prices and profit margins during the pandemic to achieve high net profit figures (Konczal and Lusiani 2022; Weber and Wasner 2023). Consistent with the idea that a firm's profitability can be persistent and help build momentum, the lagged value of RoA is also positive and highly significant.

Column (3) in Table 4 reports the results from Equation 4 which investigates whether R&D intensity moderates the influence of

AI innovation on firm profitability. Firms with higher R&D intensity attain a 0.75% increase in the next year's RoA through AI innovation compared to the nominal increase observed in Column (2), confirming that investments in aligned organizational capabilities are necessary to effectively integrate new technology into firm operations. Thus, we find support for our first and second hypotheses.

4.2 | AI Innovation, Sustainability, and Firm Value

For the third hypothesis, we investigate whether AI innovation affects firm value, particularly for firms that perform better on sustainability-related metrics. This impact is analyzed in three ways: (1) on the entire sample; (2) on a subsample with ESG score > 50; and (3) on a subsample with ESG score ≤ 50. Since we have focused on the environmental impact of AI innovation, we also use the environmental pillar score (EPS) of the ESG data as a subsampling criterion. ESG scores below 50 indicate poor performance, while scores above 50 are considered satisfactory. LSEG uses a percentile-ranked system for its ESG scoring, with a score of 50 indicating that the firm has performed better than 50% of its peers in the same industry, making 50 a reasonable threshold for separating more sustainable firms from less sustainable ones (LSEG 2024). Furthermore, the median values for the ESG and EPS scores in our sample are 55.04 and 49.52, respectively, and Table 1 shows that the mean values are also around 50, providing additional justification for choosing 50 as the cutoff value. It might also not be prudent to expect a similar market perception regarding

TABLE 4 | Regression results for the effect of AI innovation on RoA of S&P 500 companies.

Variable	(1) RoA _{i,t}	(2) RoA _{i,t}	(3) RoA _{i,t}
Constant	0.0796*** (82.05)	0.1057*** (4.24)	−1.1416*** (−3.13)
AI Patents _{i,t-1}	0.000027** (1.96)	0.000021** (1.97)	−0.00004** (−1.96)
AI Patents × R&D _{i,t-1}		—	0.0075*** (3.45)
Size _{i,t-1}		−0.0062*** (−4.34)	−0.0066*** (−4.63)
Cash Holdings _{i,t-1}		0.00082 (0.92)	0.0017 (0.20)
Leverage _{i,t-1}		0.0536*** (9.57)	0.054*** (9.66)
R&D _{i,t-1}		0.2308*** (7.77)	0.2117*** (7.01)
Profitability _{i,t-1}		0.5342*** (41.33)	0.5327*** (41.21)
COVID		0.0069*** (2.95)	0.0071*** (3.01)
DotCom		−0.0227*** (−7.79)	−0.0229*** (−7.83)
GR		−0.0254*** (−9.22)	−0.0254*** (−9.24)
RoA _{i,t-1}		0.1171*** (11.24)	0.116*** (11.22)
Industry fixed effects	No	Yes	Yes
Time fixed effects	No	Yes	Yes
N	9658	9535	9535
Adj. R ² (%)	0.04	38.12	38.20

***, **, and * imply the significance of the coefficients at 1%, 5%, and 10%, respectively. The *t*-statistics are reported in the parentheses.

Source: Authors' own compilation using Equations 3 and 4.

sustainability throughout the entire sample. Media attention to sustainability has increased significantly since the 2015 Paris Agreement, in turn increasing the focus and discussion on corporate sustainability (Zheng et al. 2024). Figure 1 shows the number of AI-related patents granted to the S&P 500 firms over the years. The patent count is also significantly higher after 2015. Hence, we perform the same analysis for two different time periods: 2000–2015 and 2016–2024. The findings are presented in Table 5, and Tables 6a and 6b. In Table 5, Columns (2), (3), and (5) are of particular interest for our study. We observe that while overall, AI innovation does not significantly impact firm value, there is a small but significant positive impact when sustainable business operations support the innovation. Tobin's Q value rises by 0.0005 with each additional AI patent. While the individual impact may seem small, its economic magnitude can be assessed by considering a realistic change in AI innovation. An increase in AI patents by one standard deviation (115.66 from Table 1) leads to the Tobin's Q value of the firm rising by 0.058, translating to a 3% increase in firm value compared to the sample mean of 1.89. This positive effect is, however, weakly significant at just the 10% level. Moving to Table 6a, we observe that between 2000 and 2015, firm value is negatively impacted by AI innovation irrespective of sustainability performance. Columns (2), (3), and (5) report that a standard deviation rise in AI patents leads to an approximate 6% fall in firm value relative to the sample mean, aligning the results with Lui et al. (2022). Huge investment costs may have led investors to have a negative perception of the firm's value. In Columns (3) and (5) of Table 6b, AI innovation in firms with better sustainable performance leads to a 6% relative increase in Tobin's Q for AI patents increasing by one standard deviation, with the effect being significant at 5% for the ESG score parameter and at 1% for the EPS parameter.

These results underscore the importance of the environmental impact associated with AI and how the markets reward firms that work towards mitigating it through sustainable

performance. It is also worth mentioning that firms that do not perform well on the sustainability parameters are not penalized by the market. The results support our third hypothesis, wherein firms focusing on sustainability, particularly environmental sustainability, experience an increase in firm value with AI innovation, while other firms experience no effect on firm value. This effect is more significant post-2015, signifying the increased attention being paid to sustainability since the 2015 Paris Agreement. As an additional robustness test for the post-2015 period, we also conducted a subsample analysis based on ESG and EPS score quartiles: Q1 (0–25), Q2 (26–50), Q3 (51–75), and Q4 (76–100). The results remain consistent with our main model: AI innovation has no significant effect on firm value in Q1 and Q2, and a positive impact in Q3 and Q4. The results are presented in Table A1.

4.3 | Impact of Political Administration

For the fourth hypothesis, we aim to investigate whether market perceptions of AI innovation and sustainability differ across political administrations. Similar to the third hypothesis, we run the regressions for the three time periods, but only for the subsample with better sustainability performance. Tables 7a and 7b display the findings. The positive results in Table 7a, with a 1% level of significance, show that firms with higher ESG and EPS scores exhibit an approximate 8% relative increase in firm value for each standard deviation increase in AI patents during Democratic regimes. This aligns with the notion that Democratic policies and their supporters place a high degree of importance on environmental issues. In Table 7b, it is noted that AI innovation has an insignificant impact on firm value during Republican administrations. This result is understandable, given that the United States announced its exit from the Paris Agreement when the Republican administration took office in 2017 and later rejoined in 2021, only after a Democratic administration was elected. The negative impact during 2000–2015 is consistent with our results in Table 6a. These results align with



FIGURE 1 | AI-related patents granted to the current S&P 500 firms over the years.

TABLE 5 | Regression results measuring the impact of AI innovation on firm value based on sustainability performance metrics.

Variable	Panel A: Full Sample		Panel B: ESG Score		Panel C: Environmental Pillar Score (EPS)	
	(1)	(2)	(3)	(4)	(5)	(6)
	Tobin's $Q_{i,t}$	Tobin's $Q_{i,t}$	Tobin's $Q_{i,t}$ ESG > 50	Tobin's $Q_{i,t}$ ESG ≤ 50	Tobin's $Q_{i,t}$ EPS > 50	Tobin's $Q_{i,t}$ EPS ≤ 50
Constant	2.1262*** (121.23)	3.611*** (7.01)	3.735*** (5.68)	7.938*** (5.74)	3.4529*** (4.78)	7.3412*** (7.6)
AI Patents $_{i,t-1}$	-0.00044* (-1.73)	0.00029 (1.22)	0.00051* (1.85)	-0.00221 (-1.46)	0.00056* (1.89)	0.00046 (0.76)
RoE $_{i,t-1}$		0.0024 (1.19)	0.0029 (1.38)	0.0039 (0.39)	0.0029 (1.34)	0.0045 (0.46)
Size $_{i,t-1}$		-0.1598*** (-5.31)	-0.1686*** (-4.41)	-0.387*** (-6.59)	-0.1597*** (-3.82)	-0.3572*** (-7.05)
Cash Holdings $_{i,t-1}$		0.2748* (1.69)	0.0449 (0.21)	0.6474*** (2.69)	0.1171 (0.49)	0.6281*** (3)
Leverage $_{i,t-1}$		0.1696 (1.2)	0.0654 (0.35)	0.4722** (2.14)	0.1534 (0.75)	0.3555* (1.87)
R&D $_{i,t-1}$		5.188*** (8.52)	6.236*** (7.6)	-2.698*** (-2.76)	5.8808*** (6.59)	-1.9117** (-2.07)
Profitability $_{i,t-1}$		4.2292*** (20.97)	4.3639*** (16)	0.6671** (2.17)	4.6045*** (15.56)	0.7491*** (2.65)
Analyst $_{i,t-1}$		0.0081*** (2.79)	0.012*** (3.2)	-0.0112** (-2.42)	0.0175*** (4.35)	-0.0065 (-1.58)
COVID		-0.5771*** (-6.86)	-0.5347*** (-6.05)	-0.7739 (-0.8)	-0.5343*** (-5.63)	-0.8133* (-1.89)
DotCom		-0.678*** (-6.13)	-0.566*** (-4.69)	-1.3522 (-1.4)	-0.4776*** (-3.76)	-1.2618*** (-2.88)
GR		-0.8129*** (-8.47)	-0.8699*** (-7.14)	-0.8969 (-0.94)	-0.8489*** (-6.66)	-0.8924** (-2.07)
Tobin's $Q_{i,t-1}$		0.3117*** (44.61)	0.2732*** (34.41)	0.4798*** (26.98)	0.2522*** (30.11)	0.4924*** (31.85)
Industry fixed effects	No	Yes	Yes	Yes	Yes	Yes
Time fixed effects	No	Yes	Yes	Yes	Yes	Yes
N	9451	8647	5708	2939	5184	3463
Adj. R^2 (%)	0.03	35.42	33.90	39.91	31.62	41.39

***, **, and * imply the significance of the coefficients at 1%, 5%, and 10%, respectively. The *t*-statistics are reported in the parentheses.

Source: Authors' own compilation using Equation 5.

TABLE 6a | Regression results measuring the impact of AI innovation on firm value based on sustainability performance metrics (2000–2015).

Variable	Panel A: Overall Sample		Panel B: ESG Score		Panel C: Environmental Pillar Score (EPS)	
	(1)	(2)	(3)	(4)	(5)	(6)
	Tobin's $Q_{i,t}$	Tobin's $Q_{i,t}$	Tobin's $Q_{i,t}$ ESG > 50	Tobin's $Q_{i,t}$ ESG ≤ 50	Tobin's $Q_{i,t}$ EPS > 50	Tobin's $Q_{i,t}$ EPS ≤ 50
Constant	1.91*** (97.45)	8.9136*** (11.66)	11.6867*** (9.98)	11.2125*** (9.55)	12.0596*** (9.57)	11.411*** (10.39)
AI Patents $_{i,t-1}$	-0.0034*** (-7.31)	-0.0013*** (-2.9)	-0.001** (-2.01)	-0.0035* (-1.88)	-0.0012** (-2.14)	-0.0003 (-0.46)
RoE $_{i,t-1}$		-0.0017 (-0.19)	0.0059 (0.46)	-0.0232** (-2.07)	0.0064 (0.49)	-0.0214* (-1.94)
Size $_{i,t-1}$		-0.4661*** (-9.96)	-0.6089*** (-8.55)	-0.599*** (-8.44)	-0.6372*** (-8.27)	-0.6142*** (-9.26)
Cash Holdings $_{i,t-1}$		0.3879* (1.84)	-0.5399 (-1.61)	0.376 (1.48)	-0.4394 (-1.21)	0.5191** (2.14)
Leverage $_{i,t-1}$		-0.8719*** (-4.34)	-1.6672*** (-5.33)	0.3639 (1.39)	-1.7594*** (-5.32)	0.3643 (1.49)
R&D $_{i,t-1}$		4.7599*** (5.44)	3.7553*** (2.68)	-2.447** (-2.18)	3.7856*** (2.58)	-2.3009** (-2.14)
Profitability $_{i,t-1}$		3.4361*** (13.92)	3.1766*** (7.6)	0.0498 (0.16)	3.2074*** (7.33)	0.2128 (0.73)
Analyst $_{i,t-1}$		0.0039 (0.95)	0.0045 (0.77)	-0.0101* (-1.89)	0.0103 (1.63)	-0.0127*** (-2.58)
COVID		Omitted	Omitted	Omitted	Omitted	Omitted
DotCom		-0.3916*** (-3.68)	-0.4163*** (-3.02)	-1.0189*** (-7.61)	-0.4211*** (-2.9)	-1.0257*** (-8.31)
GR		-0.6311*** (-7.19)	-0.7489*** (-5.84)	-0.5543*** (-5.2)	-0.7417*** (-5.51)	-0.5639*** (-5.71)
Tobin's $Q_{i,t-1}$		0.1409*** (18.53)	0.0832*** (8.25)	0.3525*** (15.93)	0.0782*** (7.6)	0.3732*** (17.35)
Industry fixed effects	No	Yes	Yes	Yes	Yes	Yes
Time fixed effects	No	Yes	Yes	Yes	Yes	Yes
N	5297	4807	2701	2106	2557	2250
Adj. R^2 (%)	1.08	26.26	24.03	33.66	24.02	35.92

***, **, and * imply the significance of the coefficients at 1%, 5%, and 10%, respectively. The *t*-statistics are reported in the parentheses.
Source: Authors' own compilation using Equation 5.

TABLE 6b | Regression results measuring the impact of AI innovation on firm value based on sustainability performance metrics (2016–2024).

Variable	Panel A: Overall Sample		Panel B: ESG Score		Panel C: Environmental Pillar Score (EPS)	
	(1)	(2)	(3)	(4)	(5)	(6)
	Tobin's $Q_{i,t}$	Tobin's $Q_{i,t}$	Tobin's $Q_{i,t}$ ESG > 50	Tobin's $Q_{i,t}$ ESG ≤ 50	Tobin's $Q_{i,t}$ EPS > 50	Tobin's $Q_{i,t}$ EPS ≤ 50
Constant	2.3551*** (74.51)	−0.9167 (−0.84)	0.208 (0.16)	0.4307 (0.15)	−1.9326 (−1.37)	0.5185 (0.23)
AI Patents $_{i,t-1}$	0.0023*** (3.56)	0.0006 (1.16)	0.001** (2.31)	−0.0044 (−0.97)	0.0011*** (2.57)	−0.0011 (−0.41)
RoE $_{i,t-1}$		0.0029 (1.51)	0.0032* (1.92)	0.0312 (1.47)	0.0032* (1.82)	0.0295 (1.56)
Size $_{i,t-1}$		0.0686 (1.04)	−0.0028 (−0.04)	0.0314 (0.17)	0.1113 (1.35)	0.0173 (0.12)
Cash Holdings $_{i,t-1}$		−0.0648 (−0.24)	−0.5785** (−2.11)	1.5347** (2.11)	−0.4882 (−1.47)	0.8961* (1.78)
Leverage $_{i,t-1}$		0.7656*** (3.05)	0.853*** (3.16)	0.5828 (0.92)	1.1365*** (3.8)	0.4541 (0.95)
R&D $_{i,t-1}$		1.6758 (1.44)	4.6299*** (3.43)	−4.4623* (−1.73)	4.0215*** (2.7)	−1.0792 (−0.5)
Profitability $_{i,t-1}$		2.3271*** (5.86)	1.8514*** (4.41)	2.8471*** (2.7)	2.4976*** (5.13)	2.2807*** (2.92)
Analyst $_{i,t-1}$		0.0018 (0.41)	0.0042 (0.94)	−0.0137 (−1.17)	0.0055 (1.13)	0.0022 (0.25)
COVID		−0.3685*** (−4.25)	−0.2766*** (−3.15)	−0.6407** (−2.23)	−0.307*** (−3.14)	−0.3931** (−2)
DotCom		Omitted	Omitted	Omitted	Omitted	Omitted
GR		Omitted	Omitted	Omitted	Omitted	Omitted
Tobin's $Q_{i,t-1}$		0.5753*** (35.93)	0.5324*** (28.01)	0.5346*** (14.31)	0.5483*** (25.77)	0.4507*** (16.16)
Industry fixed effects	No	Yes	Yes	Yes	Yes	Yes
Time fixed effects	No	Yes	Yes	Yes	Yes	Yes
N	4154	3840	3007	833	2627	1213
Adj. R ² (%)	0.34	38.52	36.16	37.70	36.17	34.35

***, **, and * imply the significance of the coefficients at 1%, 5%, and 10%, respectively. The *t*-statistics are reported in the parentheses.
Source: Authors' own compilation using Equation 5.

TABLE 7a | Regression results for AI innovation impact on firm value based on sustainability performance metrics under Democratic administration.

Variable	Panel A: 2000–2024		Panel B: 2000–2015		Panel C: 2016–2024	
	(1)	(2)	(3)	(4)	(5)	(6)
	Tobin's $Q_{i,t}$ ESG > 50	Tobin's $Q_{i,t}$ EPS > 50	Tobin's $Q_{i,t}$ ESG > 50	Tobin's $Q_{i,t}$ EPS > 50	Tobin's $Q_{i,t}$ ESG > 50	Tobin's $Q_{i,t}$ EPS > 50
Constant	3.0527*** (2.94)	3.5718*** (3.21)	4.1109*** (2.91)	3.3115** (2.1)	−1.3237 (−0.73)	−1.6616 (−0.84)
AI Patents _{$t_{i,t-1}$}	0.0013*** (2.95)	0.0013*** (2.93)	0.0016*** (3.18)	0.0015*** (2.97)	0.001 (1.63)	0.0014*** (2.11)
RoE _{$t_{i,t-1}$}	0.0043 (1.29)	0.0045 (1.29)	0.0098 (0.88)	0.0108 (0.95)	0.0032 (1.14)	0.0037 (1.23)
Size _{$t_{i,t-1}$}	−0.1502** (−2.48)	−0.1854*** (−2.87)	−0.2488*** (−2.96)	−0.2013** (−2.15)	0.0863 (0.8)	0.1054 (0.9)
Cash Holdings _{$t_{i,t-1}$}	−0.5086 (−1.56)	−0.4756 (−1.31)	−0.3203 (−0.83)	0.196 (0.45)	−0.5277 (−1.27)	−0.6791 (−1.35)
Leverage _{$t_{i,t-1}$}	0.317 (1.09)	0.3254 (1.03)	0.1239 (0.31)	−0.2377 (−0.53)	1.0251*** (2.58)	1.1743*** (2.65)
R&D _{$t_{i,t-1}$}	6.3344*** (4.8)	5.5722*** (4.06)	5.2338*** (3.03)	7.549*** (3.99)	1.9944 (1.02)	0.3284 (0.16)
Profitability _{$t_{i,t-1}$}	6.2314*** (14.71)	6.4546*** (14.19)	2.8253*** (5.05)	2.694*** (4.48)	2.8104*** (4.73)	3.0753*** (4.68)
Analyst _{$t_{i,t-1}$}	0.0159*** (2.86)	0.02*** (3.38)	0.0163*** (2.69)	0.0176*** (2.76)	0.0058 (0.81)	0.004 (0.51)
COVID	−0.4954*** (−5.21)	−0.4931*** (−4.8)	Omitted	Omitted	−0.2971*** (−2.8)	−0.3054*** (−2.58)
DotCom	−0.362** (−2.35)	−0.3641** (−2.25)	−0.1945 (−1.5)	−0.1581 (−1.15)	Omitted	Omitted
GR	−0.7924*** (−5.67)	−0.7896*** (−5.36)	−0.2211** (−2.11)	−0.1863* (−1.68)	Omitted	Omitted
Tobin's $Q_{i,t-1}$	0.2664*** (26.46)	0.2585*** (24.24)	0.5347*** (28.56)	0.5248*** (27.01)	0.5045*** (20.01)	0.5218*** (18.33)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
N	3298	3011	1426	1343	1872	1668
Adj. R^2 (%)	34.37	33.09	60.13	60.58	36.53	35.34

***, **, and * imply the significance of the coefficients at 1%, 5%, and 10%, respectively. The *t*-statistics are reported in the parentheses.

Source: Authors' own compilation using Equation 5.

TABLE 7b | Regression results for AI innovation impact on firm value based on sustainability performance metrics under Republican administration.

Variable	Panel A: 2000–2024		Panel B: 2000–2015		Panel C: 2016–2024	
	(1)	(2)	(3)	(4)	(5)	(6)
	Tobin's $Q_{i,t}$ ESG > 50	Tobin's $Q_{i,t}$ EPS > 50	Tobin's $Q_{i,t}$ ESG > 50	Tobin's $Q_{i,t}$ EPS > 50	Tobin's $Q_{i,t}$ ESG > 50	Tobin's $Q_{i,t}$ EPS > 50
Constant	4.0723*** (4.96)	3.4649*** (3.77)	8.9955*** (5.27)	9.0134*** (5.09)	-5.248** (-2.23)	-7.3335*** (-2.68)
AI Patents _{$t_{i,t-1}$}	-0.0002 (-0.68)	-0.0003 (-0.98)	-0.0013** (-2.34)	-0.002*** (-2.62)	0.0004 (0.62)	0.0004 (0.65)
RoE _{$t_{i,t-1}$}	0.0051** (2.07)	0.0055** (2.18)	-0.0226 (-0.4)	-0.0196 (-0.34)	0.0204*** (7.27)	0.0195*** (6.57)
Size _{$t_{i,t-1}$}	-0.1928*** (-3.92)	-0.1723*** (-3.13)	-0.4843*** (-4.46)	-0.4871*** (-4.28)	0.3596*** (2.67)	0.4474*** (2.89)
Cash Holdings _{$t_{i,t-1}$}	0.312 (1.14)	0.2754 (0.94)	0.8297** (1.99)	1.2605*** (2.9)	-0.5781 (-1.59)	-0.0722 (-0.15)
Leverage _{$t_{i,t-1}$}	0.1739 (0.75)	0.5462** (2.24)	-0.9588*** (-2.59)	-0.8177** (-2.16)	0.5207 (1.15)	0.787 (1.52)
R&D _{$t_{i,t-1}$}	5.557*** (5.54)	6.4281*** (5.24)	0.2363 (0.13)	-2.0974 (-1.13)	-1.3111 (-0.49)	-0.6486 (-0.2)
Profitability _{$t_{i,t-1}$}	1.4855*** (4.37)	0.9873*** (2.72)	0.7557 (1.53)	0.7015 (1.36)	-0.9144 (-1.03)	0.0206 (0.02)
Analyst _{$t_{i,t-1}$}	0.003 (0.63)	0.0058 (1.18)	-0.0024 (-0.29)	0 (-0.01)	0.0042 (0.75)	0.0132** (2.09)
COVID	0.1159 (1.54)	0.1276 (1.61)	Omitted	Omitted	0.0002 (0)	-0.0099 (-0.14)
DotCom	-0.6446*** (-6.18)	-0.564*** (-5.16)	-0.1841* (-1.66)	-0.1436 (-1.25)	Omitted	Omitted
GR	-1.1213*** (-10.96)	-1.061*** (-10.14)	-0.5836*** (-6.02)	-0.6121*** (-6.11)	Omitted	Omitted
Tobin's $Q_{i,t-1}$	0.4485*** (27.06)	0.493*** (24.66)	0.2372*** (9.45)	0.194*** (7.43)	0.6421*** (14.2)	0.6652*** (12.7)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
N	2410	2173	1275	1214	1135	959
Adj. R^2 (%)	43.29	41.90	28.95	26.47	34.70	34.25

***, **, and * imply the significance of the coefficients at 1%, 5%, and 10%, respectively. The *t*-statistics are reported in the parentheses.

Source: Authors' own compilation using Equation 5.

our fourth hypothesis that AI innovation in a sustainable firm is viewed more favorably by the market when the administration is more supportive of sustainability as a principle. This result can be understood from the perspective of institutional theory (North 1990). Institutions, defined as the “formal and informal rules of the game,” influence firm outcomes and market valuations. Managers also seek legitimacy by aligning firm decisions with existing institutional environments (Suchman 1995). Institutional investors also drive a firm’s sustainable performance when these issues find support among the community (Dyck et al. 2019). Hence, stronger institutional alignment with sustainability in Democratic regimes, through proactive environmental regulations and green incentives, could improve the legitimacy of sustainable firms and enhance market perception of their AI innovation.

Our study hence finds evidence to support H1, H2, H3, and H4. These results underscore the importance of sustainability in corporate strategic decision-making regarding AI investment. The competitive advantage of AI for businesses, in the form of higher resource utilization efficiency, process automation, and reduced human errors, cannot be denied. The results of the first and second hypotheses indicate the potential of reaping long-term benefits from continued innovation, especially for R&D-intensive firms. The third hypothesis underlines the increased attention being paid to sustainability, especially in the post-2015 period. The environmental repercussions of AI are significant, and a company’s efforts to mitigate them with a sincere focus on environmental sustainability are being perceived positively by investors and the market. While the poor performers are not penalized, the incremental reward provided to the better performers highlights the need for firms to accord sustainability its rightful place in firm-level strategic decisions regarding AI investment. To strengthen the validity of our results, we now proceed with the robustness tests in the next section.

5 | Robustness Check and Further Analysis

5.1 | Endogeneity

Our model creates a dynamic panel data set by using lagged values of the dependent variables, which can introduce endogeneity in the sample. Thus, our dynamic panel regression uses System-GMM (generalized method of moments) estimators to address these concerns (Blundell and Bond 1998). Clustered standard errors are employed in the two-step GMM model to address concerns about heteroskedasticity and autocorrelation. The results are shown in Table 8, with Columns (1)–(8) displaying the main results of interest in our study for the entire sample period. AI innovation remains statistically significant, aligning with our main regression results and indicating that our findings are robust to potential concerns about endogeneity and heteroskedasticity. AI innovation improves firm performance and enhances firm value when coupled with sustainable business practices.

5.2 | Propensity Score Matching

As a robustness check to support our panel regression and GMM estimates, we use propensity score matching (PSM). This

technique alleviates concerns about non-random selection into treatment by making treated and control firms observationally similar on important firm-level covariates. We define treated firms as those in the top quartile for the number of AI patents, signifying high innovation intensity, while the remaining firms form the control group, and then calculate the average treatment effect on the treated (ATT). Table 9 reports the results of this analysis. Panel A focuses on the treatment effect on our main outcome variables, while Panel B analyzes the covariate balancing after matching. The mean bias is significantly reduced, the pseudo- R^2 remains below 0.05, and the percentage of covariates with variance ratios between 0.5 and 2 remains above 60% for all our scenarios. This indicates that the matching of treated and control groups has been successful. The results from Panel A demonstrate that the RoA of high AI innovation intensity firms is higher by 0.5%, significant at the 10% level. This is in line with H1, albeit at a lower significance level. Hence, we obtain mixed results concerning AI innovation and its effect on firm financial performance. As far as firm value is concerned, firms with the most AI patents tend to have a Tobin’s Q premium of approximately 0.15, which is significant at the 5% level. This premium increases to 0.27 and 0.23 for firms having ESG and the environmental pillar scores above 50. This again underscores that firm value is enhanced by sustainable performance. The premium in firm value remains significant for firms with an environmental pillar score above 50 under both Democratic and Republican governments; however, the magnitude of the difference is approximately twice as large under Democratic regimes. For firms with an ESG score greater than 50, the significant positive difference in firm value is significant only during Democratic administrations. These results provide empirical support for all our proposed hypotheses.

While we have employed GMM and PSM in our study, residual endogeneity concerns remain. Since firms with higher market capitalizations may have sufficient capital and technological capabilities to invest in AI innovation, which can influence the AI-performance relationship, reverse causality cannot be fully ruled out. Time-varying omitted variables, such as shifts in managerial strategy or firm-level risk tolerance, can influence both AI innovation and market outcomes and cannot be fully resolved with fixed effects. PSM cannot account for unobservable differences between high and low-innovation firms, leading to residual selection bias. We acknowledge these residual concerns as limitations of our study which can be improved upon in future research.

5.3 | Alternative Model of Analysis—Log of AI-Patents and Interaction Terms

In our study, we use subsample analysis to examine the differential effect of AI innovation on firm value, conditional on the firm’s sustainability score. We use a numerical count of patents, which is usually considered a reliable and acceptable proxy for corporate innovation (Pham et al. 2024). This is also relevant, as we previously mentioned that training a single AI model entails significant carbon emissions. To account for the whole sample, we will need interaction terms. In this case, interacting the absolute patent count with ESG and EPS scores can cause skewness in our data. We take the natural logarithm of the number

TABLE 8 | Two-step system GMM estimation showcasing the impact of AI innovation on firm profitability and firm value.

Variable	Panel A: impact on firm profitability		Panel B: Impact on firm profitability moderated by R&D intensity		Panel C: Impact on firm value (ESG score > 50)		Panel D: Impact on firm value (EPS > 50)		Panel E: Impact on firm value under Dem leadership (ESG > 50)		Panel F: Impact on firm value under Dem leadership (EPS > 50)		Panel G: Impact on firm value under Rep leadership (ESG > 50)		Panel H: Impact on firm value under Rep leadership (EPS > 50)	
	(1) RoA _{<i>i,t</i>}	(2) RoA _{<i>i,t</i>}	(3) Tobin's Q _{<i>i,t</i>}	(4) Tobin's Q _{<i>i,t</i>}	(5) Tobin's Q _{<i>i,t</i>}	(6) Tobin's Q _{<i>i,t</i>}	(7) Tobin's Q _{<i>i,t</i>}	(8) Tobin's Q _{<i>i,t</i>}								
Constant	-0.9842 (-1.82)	-2.7457** (-2.43)	-83.6024 (-1.21)	-77.9918 (-1.21)	-71.7362*** (-2.82)	-76.129*** (-3.52)	-12.9485 (-1)	-5.0351 (-0.41)								
AI Patents _{<i>i,t-1</i>}	0.000029*** (4.61)	-0.000007 (-0.60)	0.00055** (2.12)	0.00045** (2.42)	0.00047*** (5.18)	0.00044*** (4.8)	0.00032*** (2.84)	0.00032 (1.61)								
AI Patents × R&D _{<i>i,t-1</i>}	—	0.00041** (2.04)	—	—	—	—	—	—								
RoE _{<i>i,t-1</i>}	—	—	0.00004*** (4.68)	0.00004*** (4)	0.00004* (1.95)	0.00004* (1.89)	0.00006*** (6.89)	0.00005*** (6.54)								
Size _{<i>i,t-1</i>}	-0.0051*** (-2.69)	0.0003 (0.09)	-0.3979 (-1.62)	-0.378* (-1.73)	-0.2958*** (-4.23)	-0.3017*** (-6.76)	-0.1967*** (-4.41)	-0.1714*** (-3.83)								
Cash Holdings _{<i>i,t-1</i>}	0.0024 (0.16)	0.0321* (1.69)	2.0473 (0.7)	1.9923 (0.69)	0.8213 (0.73)	0.9862 (0.94)	1.0137** (2.15)	0.5349 (0.81)								
Leverage _{<i>i,t-1</i>}	0.0775 (1.75)	0.0171 (0.95)	-0.3751 (-0.63)	-0.173 (-0.32)	0.0151 (0.06)	0.2278 (1.04)	5.5003** (2.25)	4.6554 (1.37)								
R&D _{<i>i,t-1</i>}	0.0367*** (3.25)	0.0507 (1.07)	8.9569*** (2.64)	9.1471*** (2.94)	8.5913*** (3.95)	9.1219*** (4.92)	-0.266 (-1.35)	-0.1563 (-0.66)								
Profitability _{<i>t-1</i>}	0.3983*** (2.71)	1.1071** (2.41)	5.0473 (1.13)	4.9034 (1.2)	2.7786* (1.66)	3.3416** (2.44)	2.4556 (1.52)	1.6439 (0.89)								
Analyst _{<i>i,t-1</i>}	—	—	0.024* (1.87)	0.0262** (2.05)	0.0166*** (2.78)	0.0165*** (3.51)	0.0209** (2.51)	0.022** (2.39)								
COVID	-0.0068 (-0.83)	0.0062 (0.52)	0.1811 (0.47)	0.1481 (0.42)	-0.0999 (-0.89)	-0.1044 (-0.91)	0.1643 (0.97)	0.1349 (0.82)								
DotCom	-0.0167 (-1.6)	0.0083 (-1.13)	-0.0434 (-0.08)	-0.0722 (-0.14)	-0.1426 (-0.65)	-0.1897 (-0.9)	-0.2587*** (-4.22)	-0.2751*** (-4.13)								
GR	-0.0259*** (-8.17)	-0.0303*** (-8.32)	-0.4124** (-2.24)	-0.4079** (-2.07)	-0.0395 (-0.5)	-0.0509 (-0.83)	-0.5551*** (-3.94)	-0.5598*** (-3.87)								
Lagged dep. variable	0.1796 (0.96)	-0.6981 (-1.31)	0.0681 (0.12)	0.1001 (0.2)	0.3712** (2.17)	0.3405*** (2.99)	0.3595** (2.45)	0.4559*** (2.92)								
N	9535	9535	5708	5184	3298	3011	2410	2173								
p value: AR(1)	0.000	0.255	0.691	0.637	0.004	0.002	0.017	0.020								
p value: AR(2)	0.474	0.131	0.530	0.477	0.171	0.359	0.902	0.912								
p value: Hansen	0.155	0.383	0.829	0.772	0.520	0.701	0.250	0.236								

***, **, and * imply the significance of the coefficients at 1%, 5%, and 10%, respectively. The *t*-statistics are reported in the parentheses.
Source: Authors' own compilation using Equations 3, 4, and 5.

TABLE 9 | Propensity Score Matching: Impact of AI innovation on firm profitability and firm value.

Outcome	Panel A: Treatment Effect (ATT) after PSM				Panel B: Covariate balancing before and after matching					
	Treated	Controls	Difference	t-Stat	Mean bias (%)			Pseudo R^2		
					Unmatched	Matched	%Var	Unmatched	Matched	%Var
RoA_{<i>i,t</i>}	0.08422	0.07872	0.00550*	1.78	24.7	2.0	0.191	0.001	0.001	50
Tobin's Q (Full Sample)	2.2345	2.0862	0.1483**	2.12	28.2	1.6	0.199	0.001	0.001	75
Tobin's Q (ESG > 50)	2.1979	1.9709	0.2270***	2.65	30.6	2.3	0.209	0.002	0.002	88
Tobin's Q (EPS > 50)	2.5484	2.3061	0.2423**	2.11	36.6	7.6	0.215	0.011	0.011	88
<u>Democrat Govt</u>										
Tobin's Q (ESG > 50)	2.6452	2.2281	0.4171**	2.39	30.7	4.7	0.210	0.009	0.009	75
Tobin's Q (EPS > 50)	2.6419	2.1907	0.4512***	3.00	31.6	5.3	0.209	0.008	0.008	75
<u>Republican Govt</u>										
Tobin's Q (ESG > 50)	2.2080	2.0986	0.1094	0.92	29.7	2.1	0.217	0.002	0.002	75
Tobin's Q (EPS > 50)	2.1383	1.8857	0.2526**	2.10	31.3	2.8	0.231	0.003	0.003	75

Note: %Var signifies percentage of covariates with variance ratios between 0.5 and 2; a lower pseudo- R^2 signifies better matching and reduced covariate imbalance.

***, **, and * imply the significance of the coefficients at 1%, 5%, and 10%, respectively.

Source: Authors' own compilation.

of AI-related patents to mitigate such concerns (An et al. 2018), which is defined as:

$$\text{Log (AI Patents)}_{i,t} = \ln(1 + \text{AI Patents}_{i,t})$$

We interact this variable with the ESG and EPS scores, then retest the key results of our study. This model also allows us to use ESG and EPS scores as continuous measures rather than as subsampling criteria. The findings are presented in Table 10, with Panels A and B showing that a 1% increase in the number of AI patents granted to a firm leads to a 0.004% increase in the firm's RoA, with firms having higher R&D intensity observing a greater impact. For firm value, the results remain consistent with our main analysis. These findings, hence, lend support to H1, H2, H3, and H4.

5.4 | Generative vs. Non-Generative AI Innovation

Generative AI is gaining popularity and business interest with the release of technologies such as DALL-E, Midjourney, ChatGPT, and DeepSeek. It is a subcategory of AI that analyzes underlying patterns in existing data using advanced machine learning models to produce new content, such as images, text, or videos. It now accounts for more than 20% of all AI investment (Maslej et al. 2025). It is an extremely energy-intensive process, with a ChatGPT request using about 10 times as much energy as a simple Google search (Strubell et al. 2019). Consequently, it carries a higher environmental burden in the form of increased energy demand and a larger carbon footprint (Crawford 2024; Ding et al. 2025; Bashir et al. 2024). This makes it interesting to conduct a heterogeneity analysis to assess whether there are differences in how generative AI patents affect firm performance and market value compared with other non-generative AI patents. Generative AI patents are identified as per the keywords, CPC, and IPC codes provided by the World Intellectual Property Organization (WIPO). Non-generative AI patents are calculated as the difference between total AI patents and the generative AI patents. Contemporary innovation in generative AI began after the introduction of Generative Adversarial Networks in 2014 (World Intellectual Property Organization 2024). Hence, we conduct a comparative analysis of generative AI vs. non-generative AI innovation for the period 2015–2024 to maintain parity in observations. The results of our analysis are displayed in Tables 11a and 11b. Here, we observe that while the patents alone were unable to impact firm profitability, companies with higher R&D intensity noticed a highly significant impact of AI innovation on their RoA. The impact on firm value remains mostly consistent with the results of our main analysis for both generative and non-generative AI innovation. The only difference is that generative AI innovation by firms with an ESG score lower than 50 has a weakly significant but negative impact on firm value.

6 | Contribution, Implications, Limitations, and Future Scope

Our study significantly contributes to research on the interaction among sustainability, technology, and firm performance. AI is growing at a breakneck pace, entering almost every sphere of corporate activity. All major businesses are trying to capitalize on the potential of AI innovation to revolutionize their

operations. We selected AI innovation (proxied by AI-related patents) as our primary explanatory variable because the development of proprietary, in-house technology will be a critical factor for firms seeking to develop a strategic advantage over their competitors moving forward. Our study first provides evidence that investing in AI innovation can yield tangible profitability for firms, with more R&D-intensive firms observing better results. While this result focuses on why firms opt for AI innovation, it is equally important to examine how their actions affect the environment around them and whether the market places any emphasis on the same. AI innovation, in all its stages (development, training, and deployment), is an energy-intensive process with an enormous carbon footprint. Hence, discussing the role of sustainability in AI innovation is paramount. This study is the first to analyze how the market perceives AI innovation by sustainable firms compared to other firms. The results demonstrate that AI innovation, coupled with sustainable performance, increases firm value. However, poor sustainability performers are not penalized in terms of firm value; instead, better performers tend to receive higher market valuations as AI innovation increases. The results remain significant even after applying multiple robustness checks, including a two-step system GMM, PSM, and an alternative analysis using interaction terms and the logarithm of AI patents as independent variables. Our study also observes the impact of the political environment on this market perception. The rise in firm value is significant only during Democratic governments in the United States, aligning with institutional and legitimacy theories that highlight how the institutional policy environment can shape market perceptions of partisan topics.

The study has theoretical implications across different strategic frameworks. From the perspective of the resource-based view (Barney 1991), firms can generate a sustained competitive advantage through AI patents as they represent an inimitable, rare, and valuable organizational resource, reflected here as a positive impact on firm profitability. The dynamic capabilities framework posits that sustaining such a competitive advantage depends on developing internal technological processes within the firm (Teece et al. 1997), as evidenced by R&D intensity moderating the impact of AI innovation on firm profitability. The positive effect of AI innovation on the value of sustainable firms aligns with the value-enhancing theory (Hart and Milstein 2003), supporting the argument that financial markets reward more socially responsible firms (Malik 2015). This paper has implications for different stakeholders. Managers and executives can encourage AI innovation, coupled with an increase in complementary R&D intensity, to effectively integrate new technologies and improve the financial performance of their firms. To manage AI investments and firm sustainability goals, they should integrate sustainability metrics in their AI innovation process, like using AI to reduce a firm's carbon footprint, or developing AI with energy-efficient models and less computationally-intensive algorithms (Bolón-Canedo et al. 2024), or incorporating more renewable energy sources in their energy mix. Disclosure of such green AI innovation practices in their annual reports can strengthen market signaling and enhance firm value. Since financial markets value AI innovation by sustainable firms, policymakers can encourage sustainable innovation by providing targeted incentives, such as tax breaks and subsidies, linked to measurable green AI innovation

TABLE 10 | Regression results for impact of AI innovation on firm profitability and firm value using interaction terms.

Variable	Panel A: Impact on firm profitability	Panel B: Impact on firm profitability moderated by R&D intensity	Panel C: Impact on firm value (based on ESG score)	Panel D: Impact on firm value (based on EPS score)	Panel E: Impact on firm value under Dem leadership (based on ESG score)	Panel F: Impact on firm value under Dem leadership (based on EPS score)	Panel G: Impact on firm value under Rep leadership (based on ESG score)	Panel H: Impact on firm value under Rep leadership (based on EPS score)
	RoA _{<i>i,t</i>}	RoA _{<i>i,t</i>}	Tobin's <i>Q</i> _{<i>i,t</i>}	Tobin's <i>Q</i> _{<i>i,t</i>}	Tobin's <i>Q</i> _{<i>i,t</i>}	Tobin's <i>Q</i> _{<i>i,t</i>}	Tobin's <i>Q</i> _{<i>i,t</i>}	Tobin's <i>Q</i> _{<i>i,t</i>}
Constant	0.10623*** (4.27)	−0.9871*** (−2.69)	1.3941*** (2.69)	1.317** (2.51)	0.2407 (0.31)	0.2522 (0.32)	1.4704* (1.78)	1.3996* (1.69)
Log (AI Patents) _{<i>i,t-1</i>}	0.00451*** (3.89)	0.0026* (1.85)	−0.0698* (−1.94)	−0.0274 (−0.97)	−0.1517*** (−2.7)	−0.1254*** (−2.82)	−0.0759 (−1.46)	−0.0101 (−0.25)
Log (AIP) × R&D _{<i>i,t-1</i>}	—	0.0369*** (2.69)	—	—	—	—	—	—
ESG Score _{<i>i,t-1</i>}	—	—	−0.0005 (−0.41)	—	−0.0028 (−1.59)	—	0.0011 (0.62)	—
Log (AIP) × ESG _{<i>i,t-1</i>}	—	—	0.0013*** (2.91)	—	0.0025*** (3.27)	—	0.0016** (2.56)	—
EPS Score _{<i>i,t-1</i>}	—	—	—	−0.0007 (−0.88)	—	−0.0022* (−1.82)	—	−0.0001 (−0.08)
Log (AIP) × EPS _{<i>i,t-1</i>}	—	—	—	0.0007** (2.24)	—	0.0022*** (3.79)	—	0.0007* (1.67)
RoE _{<i>i,t-1</i>}	—	—	0.00002 (1.27)	0.00002 (1.27)	0.00002 (1.07)	0.00002 (1.05)	0.00003 (1.43)	0 (1.39)
Size _{<i>i,t-1</i>}	−0.0064*** (−4.58)	−0.0069*** (−4.81)	−0.0503 (−1.56)	−0.047 (−1.45)	0.0177 (0.39)	0.0135 (0.29)	−0.0542 (−1.06)	−0.0496 (−0.97)
Cash Holdings _{<i>i,t-1</i>}	−0.00139 (−0.17)	0.0027 (0.33)	0.1192 (0.77)	0.1127 (0.73)	−0.0486 (−0.22)	−0.0413 (−0.18)	0.4452* (1.93)	0.4351* (1.89)
Leverage _{<i>i,t-1</i>}	0.05334*** (9.55)	0.0531*** (9.48)	0.6144*** (4.65)	0.6336*** (4.81)	0.8992*** (4.8)	0.9245*** (4.96)	0.4971** (2.51)	0.518*** (2.61)
R&D _{<i>i,t-1</i>}	0.24064*** (8.18)	0.177*** (4.84)	1.12* (1.73)	1.1541* (1.78)	0.4955 (0.53)	0.4737 (0.51)	1.9867** (2.13)	1.9863** (2.13)
Profitability _{<i>t-1</i>}	0.53671*** (41.81)	0.5307*** (41.00)	1.5375*** (6.77)	1.5655*** (6.9)	2.1513*** (7.08)	2.1708*** (7.16)	0.4915 (1.3)	0.5312 (1.41)
Analyst _{<i>i,t-1</i>}	—	—	−0.0024 (−0.93)	−0.0025 (−0.96)	0.002 (0.56)	0.002 (0.57)	−0.0014 (−0.34)	−0.0014 (−0.36)
COVID	−0.00128 (−0.28)	0.0066*** (2.82)	−0.7122*** (−6.78)	−0.683*** (−6.57)	−0.7443*** (−11.37)	−0.7428*** (−11.36)	0.0643 (0.54)	0.1243 (1.05)
DotCom	−0.0265*** (−4.46)	−0.0222*** (−7.58)	Omitted	Omitted	Omitted	Omitted	Omitted	Omitted
GR	−0.04066*** (−7.91)	−0.0255*** (−9.28)	−0.1969** (−2.16)	−0.1852** (−2.02)	−0.1801* (−1.95)	−0.1799** (−1.98)	−0.97*** (−11.04)	−0.9489*** (−10.74)
Lagged dep. variable	0.1171*** (11.39)	0.1159*** (11.22)	0.6589*** (61.61)	0.6594*** (61.65)	0.6598*** (49.58)	0.659*** (49.53)	0.6608*** (32.37)	0.6618*** (32.37)
<i>N</i>	9535	9535	6992	6992	4174	4174	2818	2818
Adj. <i>R</i> ² (%)	38.19	38.24	50.10	50.07	53.02	53.06	46.93	46.80

***, **, and * imply the significance of the coefficients at 1%, 5%, and 10%, respectively. The *t*-statistics are reported in the parentheses.

Source: Authors' own compilation.

TABLE 11a | Regression results comparing the impact of generative and non-generative AI innovation on firm profitability and firm value.

Variable	Panel A: Impact on firm profitability (RoA)		Panel B: Impact on firm profitability moderated by R&D intensity (RoA)		Panel C: Impact on firm value (Tobin's Q)	
	(1)	(2)	(3)	(4)	(5)	(6)
	Gen AI	Non-Gen AI	Gen AI	Non-Gen AI	Gen AI	Non-Gen AI
Constant	-5.4262*** (-5.82)	-5.5319*** (-5.95)	-5.4613*** (-5.86)	-5.412*** (-5.97)	-5.4099*** (-3.17)	-5.6259*** (-3.31)
AI Patents _{<i>t,t-1</i>}	0.0014 (1.25)	0.00005 (1.54)	-0.00018 (-1.07)	-0.00004 (-1.04)	0.0027 (1.33)	0.0008 (1.46)
AI Patents × R&D _{<i>t,t-1</i>}	—	—	0.0049*** (2.63)	0.0018*** (3.95)	—	—
RoE _{<i>t,t-1</i>}	—	—	—	—	0.00003* (1.71)	0.00003* (1.72)
Size _{<i>t,t-1</i>}	-0.0057 (-1.59)	-0.0058 (-1.60)	-0.0063* (-1.74)	-0.0065* (-1.80)	0.0262 (0.38)	0.0257 (0.38)
Cash Holdings _{<i>t,t-1</i>}	0.0188 (1.24)	0.0176 (1.17)	0.0194 (1.28)	0.0206 (1.36)	-0.0749 (-0.27)	-0.0986 (-0.36)
Leverage _{<i>t,t-1</i>}	0.0973*** (9.19)	0.0972*** (9.19)	0.0969*** (9.16)	0.0966*** (9.14)	0.8588*** (3.32)	0.8593*** (3.32)
R&D _{<i>t,t-1</i>}	0.0542 (0.84)	0.0557 (0.86)	0.0025 (0.04)	-0.0014 (-0.02)	1.8703 (1.55)	1.8805 (1.56)
Profitability _{<i>t-1</i>}	0.5606*** (20.93)	0.5601*** (20.91)	0.5539*** (20.60)	0.5529*** (20.63)	2.0198*** (4.95)	2.0082*** (4.92)
Analyst _{<i>t,t-1</i>}	—	—	—	—	0.0034 (0.74)	0.0032 (0.7)
COVID	0.0032 (1.41)	0.0031 (1.37)	0.003 (1.35)	0.0032 (1.40)	0.0465 (1.13)	0.0449 (1.09)
Lagged dep. variable	-0.0206 (-0.97)	-0.0203 (-0.95)	-0.0219 (-1.03)	-0.0222 (-1.04)	0.5516*** (33.48)	0.552*** (33.56)
N	4160	4160	4160	4160	3840	3840
Adj. R ² (%)	21.86	21.88	22.01	22.21	33.25	33.25

***, **, and * imply the significance of the coefficients at 1%, 5%, and 10%, respectively. The *t*-statistics are reported in the parentheses.

Source: Authors' own compilation using Equations 3, 4, and 5.

TABLE 11b | Regression results comparing the impact of generative and non-generative AI innovation on firm value based on sustainability performance metrics.

Variable	Panel A: Impact on firm value based on ESG score > 50 (Tobin's Q)		Panel B: Impact on firm value based on ESG score ≤ 50 (Tobin's Q)		Panel C: Impact on firm value based on EPS score > 50 (Tobin's Q)		Panel D: Impact on firm value based on EPS score ≤ 50 (Tobin's Q)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Constant	-5.0617*** (-3.05)	-5.4530*** (-3.29)	-11.820* (-1.69)	-11.1485 (-1.59)	-3.2075* (-1.78)	-3.7179** (-2.06)	-10.3877** (-2.14)	-9.8428** (-2.03)
AI Patents _{it-1}	0.005*** (2.75)	0.0011*** (2.33)	-0.0175* (-1.69)	-0.0016 (-0.32)	0.0062*** (3.26)	0.0012** (2.42)	-0.012 (-1.58)	0.0004 (0.13)
RoE _{it-1}	0.00004** (2.21)	0.00004** (2.23)	0.0001 (0.57)	0.0001 (0.57)	0.00004** (2.09)	0.00004** (2.09)	0.0001 (0.65)	0.0001 (0.66)
Size _{it-1}	-0.0602 (-0.77)	-0.0578 (-0.74)	-0.138 (-0.73)	-0.1339 (-0.71)	0.0796 (0.93)	0.0802 (0.93)	-0.0842 (-0.58)	-0.089 (-0.61)
Cash Holdings _{it-1}	-0.5244* (-1.84)	-0.5613** (-1.97)	1.5321** (2.03)	1.6099** (2.13)	-0.3975 (-1.16)	-0.4622 (-1.35)	0.8788* (1.68)	0.9153* (1.74)
Leverage _{it-1}	1.0603*** (3.8)	1.0771*** (3.86)	0.1672 (0.25)	0.1997 (0.3)	1.3231*** (4.31)	1.3325*** (4.34)	0.3205 (0.65)	0.2879 (0.58)
R&D _{it-1}	4.7078*** (3.33)	4.6552*** (3.3)	-3.9407 (-1.46)	-4.0532 (-1.49)	4.676*** (3.03)	4.5495*** (2.94)	-1.1166 (-0.5)	-1.1632 (-0.51)
Profitability _{t-1}	1.366*** (3.15)	1.3501*** (3.11)	2.9064*** (2.68)	2.9167*** (2.67)	2.0133*** (4.04)	1.9869*** (3.98)	2.2507*** (2.79)	2.2351*** (2.76)
Analyst _{it-1}	0.0065 (1.38)	0.006 (1.28)	-0.0071 (-0.58)	-0.0095 (-0.77)	0.008 (1.58)	0.0073 (1.45)	0.0045 (0.47)	0.0037 (0.39)
COVID	0.0069 (0.18)	0.0053 (0.14)	0.2715* (1.84)	0.2694* (1.82)	0.0012 (0.03)	-0.0005 (-0.01)	0.2337** (2.28)	0.2274** (2.22)
Lagged dep. variable	0.5105*** (26.06)	0.5116*** (26.13)	0.489*** (12.73)	0.4847*** (12.59)	0.5241*** (24.14)	0.5262*** (24.22)	0.4146*** (14.41)	0.4115*** (14.31)
N	3007	3007	833	833	2627	2627	1213	1213
Adj. R ² (%)	29.76	29.70	30.51	30.21	30.90	30.75	27.26	27.08

***, **, and * imply the significance of the coefficients at 1%, 5%, and 10%, respectively. The *t*-statistics are reported in the parentheses.
Source: Authors' own compilation using Equation 5.

(e.g., energy and computationally efficient model development). They can also increase funding for research in sustainable AI development. Investors can incorporate sustainability-related AI innovation indicators into their portfolio screening models as a signal of market value. Long-term investors can sustain investment in AI innovation by sustainable firms as a value-creating strategy, since the effects of political cycles can average out over a longer investment horizon (Starks et al. 2026). Short-term investors can actively adjust portfolio allocation for such firms depending on the political regime.

However, this study is not without its limitations. Firstly, it focuses solely on the US market. While the United States is a leader in AI research and development, countries like China are also emerging as a hotbed of AI innovation. The scope of the analysis is also limited to S&P 500 firms, which can restrict the generalizability of the results. The results may be strong for large, resource-rich firms with existing technological capabilities and may not apply to smaller firms or non-US markets. Future studies can conduct a multi-country analysis to bring out a more comprehensive impact of AI innovation on firm value. They can also incorporate more firms in their sample and perform a heterogeneity analysis across industries, market capitalizations, and other firm characteristics. Using AI patents as an innovation proxy also has its limitations. Patents capture invention capacity more than adoption intensity, and the propensity to patent also varies across firms. Alternative proxies like R&D investments and AI adoption intensity (Babina et al. 2024) can be used to complement the AI patent count to achieve a more comprehensive measure of AI innovation. Future research can also conduct a comparative study between AI innovators and AI adopters. This may help shed light on how the market perceives sustainable performance in relation to active AI innovation and passive AI adoption. Investor attention could also be included as an additional explanatory variable to test if the impact of AI innovation and sustainability on firm value differs depending on the attention it receives by the investor community. Comprehensive studies on the topic can help different stakeholders adopt sustainability as a core company strategy, rather than just an obligatory afterthought.

7 | Conclusion

We analyzed the effect of AI innovation on firm financial performance and firm value. The results suggest AI innovation to be a financially profitable endeavor for firms, with firms investing in complementary capabilities, via a higher R&D intensity, being better positioned to translate this innovation into increased profitability. The results also provide evidence that the market rewards AI innovation when coupled with sustainable performance, especially post the 2015 Paris Agreement, highlighting the increasing relevance of corporate sustainability among investors. The broader institutional environment also plays an important role in this scenario, with the increase in firm value being consistently significant during Democratic governments in the United States, which is perceived as more pro-environment in its policymaking decisions.

Hence, our study has important implications for multiple groups, including investors, policymakers, and firm executives. With AI

set to become an indispensable part of business processes at a large scale, this paper makes a strong case for adopting sustainability as a fundamental pillar of corporate strategy as they invest in such technologies. It shows that sustainability and firm growth can co-exist, and firms do not necessarily have to choose between them.

Author Contributions

Anirudh Singh: conceptualization, methodology, software, data curation, formal analysis, visualization, writing – original draft, investigation, methodology, supervision, writing – review and editing, validation, resources, data curation. **Madhumita Chakraborty:** conceptualization, methodology, supervision, writing – review and editing. **Vivek Singh:** conceptualization, methodology, software, data curation, formal analysis, visualization, writing – original draft, investigation, methodology, supervision, writing – review and editing, validation, resources, data curation.

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TABLE A1 | Regression results measuring the impact of AI innovation on firm value based on sustainability performance metrics (2016–2024).

Variable	Panel A: impact on firm value based on ESG score quartiles				Panel B: impact on firm value based on EPS score quartiles			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Q1 (0–25) Tobin's $Q_{i,t}$	Q2 (26–50) Tobin's $Q_{i,t}$	Q3 (51–75) Tobin's $Q_{i,t}$	Q4 (76–100) Tobin's $Q_{i,t}$	Q1 (0–25) Tobin's $Q_{i,t}$	Q2 (26–50) Tobin's $Q_{i,t}$	Q3 (51–75) Tobin's $Q_{i,t}$	Q4 (76–100) Tobin's $Q_{i,t}$
Constant	−22.401 (−0.53)	−5.1705 (−0.78)	−9.7553*** (−3.59)	0.7477 (0.02)	−19.4882* (−1.76)	−4.0893 (−0.7)	−4.9085 (−1.18)	−2.5644 (−1.38)
AI Patents $_{i,t-1}$	0.1454 (1.5)	−0.0047 (−1.09)	0.0021** (2.25)	0.0014** (2.32)	0.0009 (0.07)	−0.0017 (−0.52)	0.0071*** (5.84)	0.00001 (0.04)
RoE $_{i,t-1}$	0.0014 (0.3)	0.0001 (0.43)	0.0001*** (2.8)	0.00003 (0.91)	0.00001 (−0.07)	0.0013* (1.89)	0.00004 (1.25)	0.00003* (1.7)
Size $_{i,t-1}$	−0.4324 (−0.45)	0.1131 (0.6)	−0.2545** (−2.23)	0.2339 (1.6)	0.0366 (0.13)	−0.1502 (−0.72)	0.0786 (0.46)	0.076 (0.81)
Cash Holdings $_{i,t-1}$	4.2193 (0.79)	1.3952*** (1.98)	−0.7061* (−1.81)	−1.3862** (−2.33)	1.4012 (1.45)	0.5847 (0.92)	−1.1814* (−1.86)	0.0335 (0.09)
Leverage $_{i,t-1}$	0.9132 (0.26)	1.2666* (1.88)	1.0059*** (2.64)	0.6549 (1.24)	0.1959 (0.23)	1.212* (1.71)	1.5411*** (2.63)	0.8824*** (2.6)
R&D $_{i,t-1}$	1.6843 (0.12)	−5.9124** (−2.01)	2.3169 (1.17)	9.4281*** (3.24)	2.621 (0.62)	−17.9556*** (−4.63)	2.5145 (0.96)	9.8761*** (4.37)
Profitability $_{i-1}$	−0.5755 (−0.11)	3.6645*** (3.47)	0.6552 (1.07)	4.2318*** (4.52)	4.1913** (2.49)	1.9303* (1.88)	1.7183* (1.88)	1.7342*** (3.08)
Analyst $_{i,t-1}$	0.1007 (0.99)	−0.0109 (−0.98)	0.011* (1.66)	0.0019 (0.23)	−0.0222 (−0.98)	0.0057 (0.56)	0.0136 (1.41)	0.0008 (0.16)
COVID	−1.3252 (−1.28)	0.3443** (2.53)	0.0755 (1.35)	−0.0035 (−0.05)	0.4097* (1.73)	0.2184** (1.98)	0.0334 (0.4)	0.0108 (0.24)
Lagged dep. variable	0.8067*** (5.77)	0.4561*** (10.1)	0.3547*** (12.79)	0.4708*** (10.8)	0.4204*** (7.85)	0.2478*** (6.01)	0.4635*** (13.35)	0.4839*** (14.56)
N	101	732	1831	1176	492	721	1319	1308
Adj. R ² (%)	50.99	31.68	17.84	29.32	30.68	23.20	28.55	31.13

***, **, and * imply the significance of the coefficients at 1%, 5%, and 10%, respectively. The *t*-statistics are reported in the parentheses.
Source: Authors' own compilation using Equation 5.